



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

AD1103-1 – SOFTWARE ENGINEERING
(AI&DS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Software is defined as**
 A) set of programs, documentation & configuration of data
 B) set of programs
 C) documentation and configuration of data
 D) None of the mentioned
2. **What is Software Engineering?**
 A) Designing a software
 B) Testing a software
 C) Application of engineering principles to the design a software
 D) None of the mentioned
3. **“Consider a system where, a heat sensor detects an intrusion and alerts the security company.” What kind of a requirement the system is providing?**
 A) Functional
 B) Non-Functional
 C) Known Requirement
 D) None of the mentioned
4. **Selection of a model is based on**
 A) Requirements
 B) Project type and associated risk
 C) Development team & Users
 D) All of the mentioned
5. **Which of the following life cycle model can be chosen if the development team has less experience on similar projects?**
 A) Spiral
 B) RAD
 C) Waterfall
 D) Iterative Enhancement Model
6. **Agile Software Development is based on**
 A) Incremental Development
 B) Linear Development
 C) Iterative Development
 D) Both Incremental and Iterative Development
7. **How is plan driven development different from agile development?**
 A) Outputs are decided through a process of negotiation during the software development process
 B) Specification, design, implementation, and testing are interleaved
 C) Iteration occurs within activities
 D) All of the mentioned
8. **How many phases are there in Scrum?**
 A) Two
 B) Three
 C) Four
 D) Scrum is an agile method which means it does not have phases
9. **Which of the following interaction models is primarily based on the idea of passing messages between different components?**
 A) Object-Oriented Model
 B) Event-Driven Model
 C) Client-Server Model
 D) Peer-to-Peer Model
10. **Which of the following is NOT a component of a Context Model in software engineering?**
 A) Environment
 B) Processes
 C) Actors
 D) Data flow

11. Which UML diagram is used to visualize the dynamic behavior of a system, including the interactions between various objects?
 - A) Class diagram
 - B) Sequence diagram
 - C) Activity diagram
 - D) Use case diagram
12. In UML, what does a solid line with an arrowhead between two classes represent?
 - A) Aggregation
 - B) Inheritance
 - C) Association
 - D) Composition
13. Which UML diagram is used to represent the static view of a system, showing the classes and their relationships?
 - A) Class diagram
 - B) Sequence diagram
 - C) Activity diagram
 - D) Use case diagram
14. Which of the following is NOT a primary goal of software testing?
 - A) Finding defects
 - B) Confirming that the software meets requirements
 - C) Ensuring 100% defect-free software
 - D) Providing information for decision-making
15. What is the main purpose of regression testing?
 - A) To verify that the system behaves as expected
 - B) To test the performance of the system under stress conditions
 - C) To ensure that new changes do not adversely affect existing functionality
 - D) To assess the usability of the software
16. Which of the following is true about validation testing?
 - A) It ensures that the software meets the specified requirements.
 - B) It focuses on finding defects in the software code.
 - C) It is performed before unit testing.
 - D) It is also known as white-box testing.
17. Which of the following best describes the purpose of risk management in software engineering?
 - A) To eliminate all risks associated with software development
 - B) To assign blame when risks lead to project failures
 - C) To identify, assess, and mitigate potential risks that could impact project success
 - D) To increase the complexity of the project by considering all possible risks
18. Which of the following scheduling techniques in software engineering involves breaking down the project into smaller tasks and estimating the time required for each task?
 - A) Critical Path Method (CPM)
 - B) Work Breakdown Structure (WBS)
 - C) Program Evaluation and Review Technique (PERT)
 - D) Gantt chart
19. Which of the following is NOT a primary goal of Quality Management in software engineering?
 - A) Ensuring customer satisfaction
 - B) Meeting project deadlines
 - C) Minimizing project cost
 - D) Delivering high-quality software products
20. In the context of software quality management, what does the acronym "ISO" stand for?
 - A) International Organization for Standardization
 - B) International Software Organization
 - C) Institute for Software Optimization
 - D) Information Security Oversight

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Briefly explain the essential attributes of good software. List out the fundamental software engineering activities.	8	L*2	1	1
	b) Explain the Waterfall model. What are the advantages and disadvantages?	8	2	1	1
2.	a) Briefly, discuss various phases of software development life cycle.	8	2	1	1
	b) What is scrum? Briefly explain the key ceremonies a scrum team might partake in.	8	2	2	1
3.	a) Explain the structure of a requirements document in requirement engineering.	8	2	2	1
	b) Explain the principles of agile methods.	8	2	2	1

Unit – II

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| 4. | a) | Briefly explain the System Testing. | 8 | 2 | 3 | 1 |
| | b) | You have been asked to develop a system that will help with planning large-scale events and parties such as weddings, graduation celebrations, birthday parties, etc. Using an activity diagram, model the process context for such a system that shows the activities involved in planning a party (booking a venue, organizing invitations, etc.) and the system elements that may be used at each stage. | 8 | 3 | 3 | 1 |
| 5. | a) | How to identify object classes in object-oriented systems? And briefly explain the Architectural design using the use case of Weather station. | 8 | 2 | 4 | 1 |
| | b) | Briefly explain the integration and smoke testing Strategies for Conventional Software. | 8 | 2 | 3 | 1 |
| 6. | a) | Briefly explain the concept of system models. | 8 | 2 | 3 | 1 |
| | b) | Explain the four fundamental architectural views. | 8 | 2 | 4 | 1 |

Unit – III

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|----|----|---|---|---|---|---|
| 7. | a) | Describe the importance of risk identification in project management. | 8 | 2 | 5 | 1 |
| | b) | Explain the concept of configuration management in software development projects. | 8 | 2 | 5 | 1 |
| 8. | a) | Explain the concept of project scheduling and its significance in project management. | 8 | 2 | 5 | 1 |
| | b) | Discuss the importance of quality management in project execution and delivery. | 8 | 2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
